

Practical No : 05

Practical Title:

Setup your own cloud for Software as a Service (SaaS) over the existing LAN in your laboratory. In this assignment you have to write your own code for cloud controller using open-source technologies to implement with HDFS. Implement the basic operations may be like to divide the file in segments/blocks and upload/ download file on/from cloud in encrypted form.

Objectives: To set your own cloud for SaaS over existing LAN

To implement the basic operations may be like to divide the file in segments/blocks

Hardware Requirements :

- Pentium IV with latest configuration

Software Requirements :

- Ubuntu 20.04, VMwareESXi cloud

Theory:

Here we are installing VMwareESXi cloud

Host/NodeESXi installation:- ESXiHardwareRequirements:-

- ESXi6.7requiresahostmachinewithatleasttwoCPUcores.
- ESXi6.7supports64-bitx86processors
- ESXi6.7requirestheNX/XDbit to be enabled for the CPU in the BIOS.
- ESXi6.7requiresaminimumof4GBofphysicalRAM.Itisrecommended to provide atleast 8 GB of RAM to run virtual machines in typical productionenvironments.
- Tosupport64-bitvirtualmachines,support for hardware virtualization (IntelVT-xor AMDRVI) mustbeenabledonx64CPUs.
- One or more Gigabit or faster Ethernet controllers. For a list of supportednetwork adapter models.
- SCSI disk oralocal,non-network,RAIDLUN with unpartitioned space for the virtualmachines ForSerialATA(SATA), a disk connected through supported SAS controller or supported on board SATA controllers. SATA disks are considered remote not local. These disks are not used as a scratch partition by default be cause they are seen as remote.

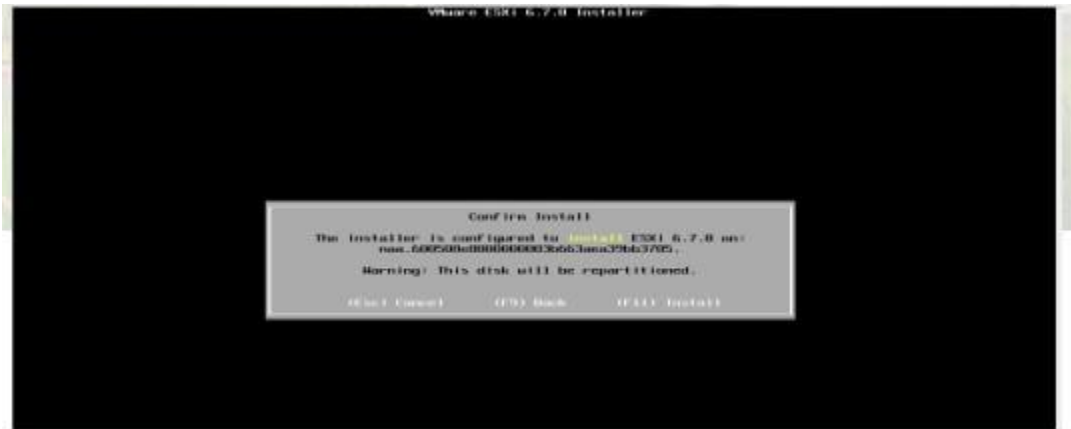
Select storage :



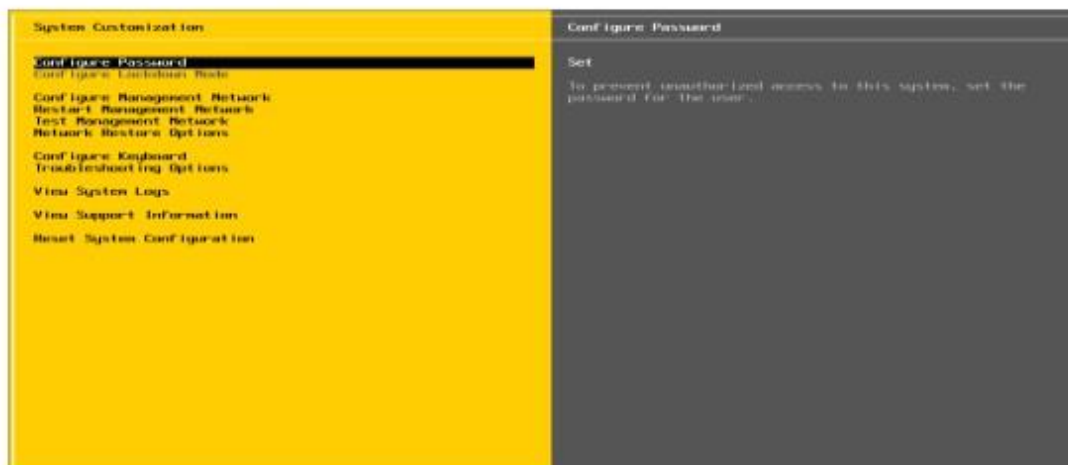
Select Keyboard Layout :



Set NodeESXi Root Password :



Installation complete (Reboot) CLI interface to configuration



CLI Interface to Configuration:



Configure Management Network



Set IPV4



Set DNServer :

Restart Management Network

Restart Management Network

System Customization	Restart Management Network
Configure Password	Restarting the management network interface may be required to restore networking or to renew a DHCP lease.
Configure Lockdown Mode	
Configure Management Network	Restarting the management network will result in a brief network outage that may temporarily affect running virtual machines.
Restart Management Network	
Test Management Network	
Network Restore Options	Note: If a renewed DHCP lease results in a new network identity (e.g., IP address or hostname), remote management software will be disconnected.
Configure Keyboard	
Troubleshooting Options	
View System Logs	
View Support Information	
Reset System Configuration	

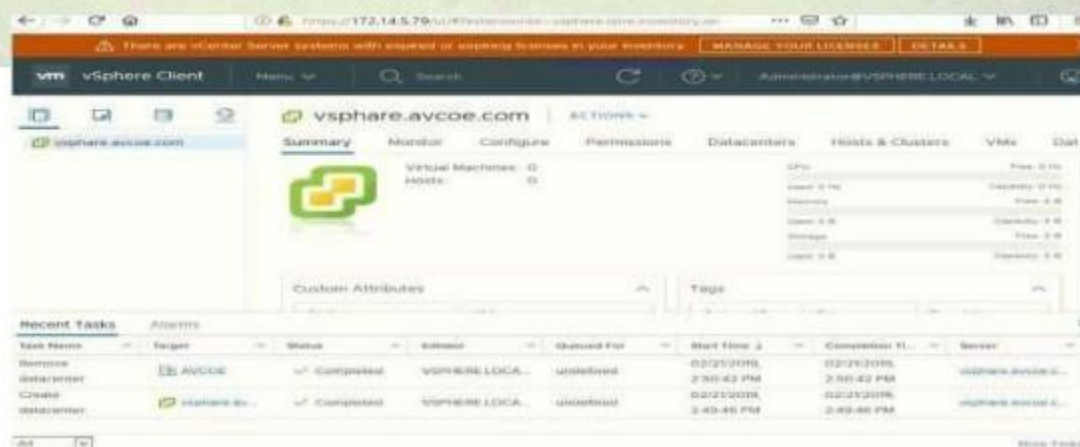
GUIAccess :



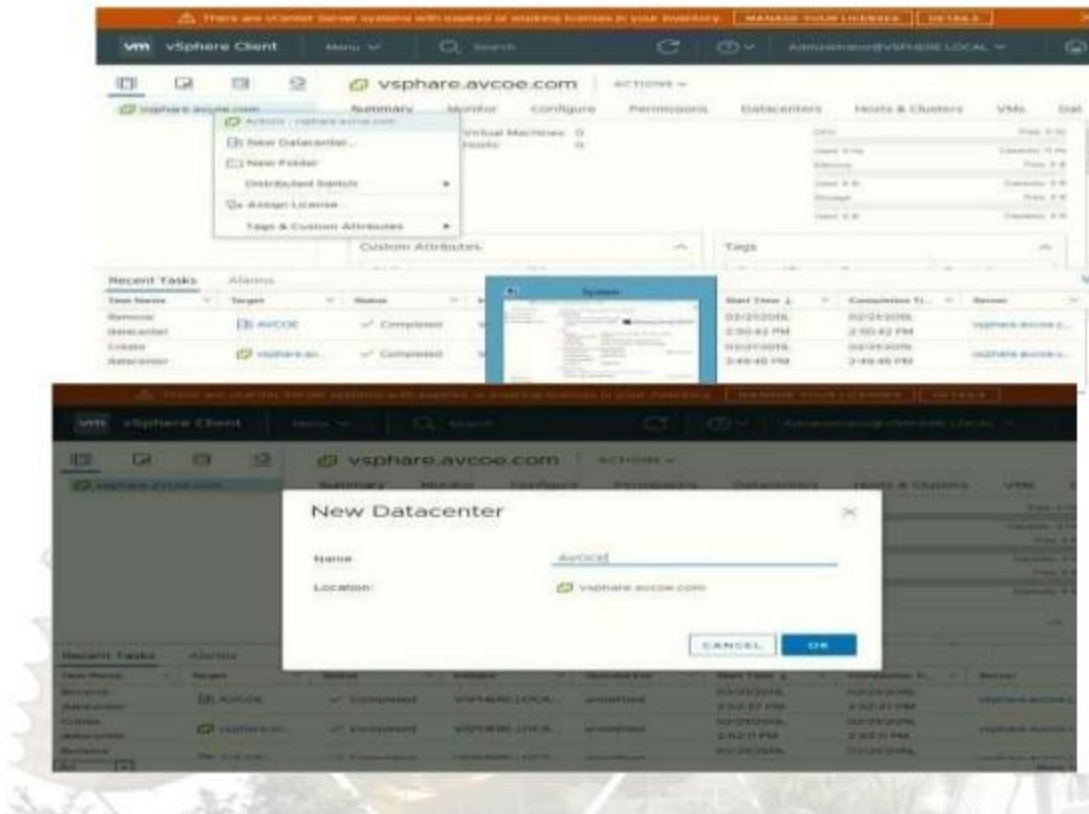
ClusterSetup

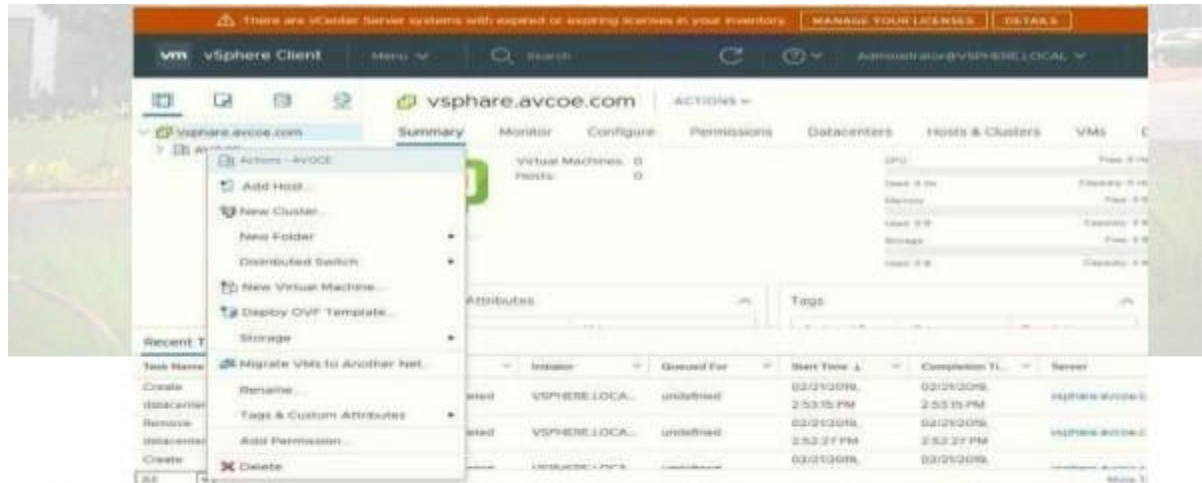
- CreatingDatacenter
- CreatingCluster
- Adding Hosts incluster
- Resourcesafteraddingcluster.
- DRS
- Failover

VCenter Access:



Create Datacenter:





Create cluster :

Assign cluster name :



Add host .:

There are vCenter Server systems with expired or expiring licenses in your inventory. [MANAGE YOUR LICENSES](#) [DETAILS](#)

vm vSphere Client Menu Search Administrator@VSPHERE.LOCAL

AVCOE-CLUSTER ACTIONS

Monitor Configure Permissions Hosts VMs Datastores Networks

Total Processors: 0
Total vMotion Migrations: 0

CPU Free: 0 Hz
Used: 0 Hz Capacity: 0 Hz
Memory Free: 0 B
Used: 0 B Capacity: 0 B
Storage Free: 0 B
Used: 0 B Capacity: 0 B

vSphere DRS

Recent Tasks

Task Name

Create cluster

Remove cluster

Provision cluster

All

Actions - AVCOE-CLUSTER

- Add Host...
- New Virtual Machine...
- New Resource Pool...
- Deploy OVF Template...
- New vApp...
- Storage
- Host Profiles
- Edit Default VM Compatibility
- Assign License...
- Settings
- Rename...
- Tags & Custom Attributes
- Delete
- vSAN

Initiator Queued For Start Time Completion Time Server

VSPHERE.LOCAL...	undefined	02/21/2019, 2:56:54 PM	02/21/2019, 2:56:55 PM	vsphere.avcoe.c...
VSPHERE.LOCAL...	undefined	02/21/2019, 2:56:01 PM	02/21/2019, 2:56:01 PM	vsphere.avcoe.c...
undefined	undefined	02/21/2019,	02/21/2019,	vsphere.avcoe.c...

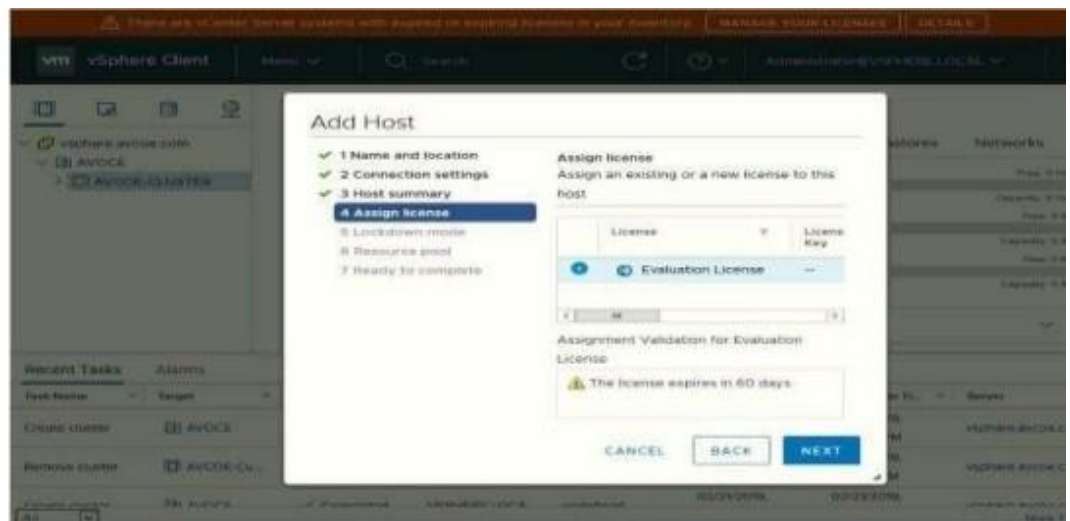
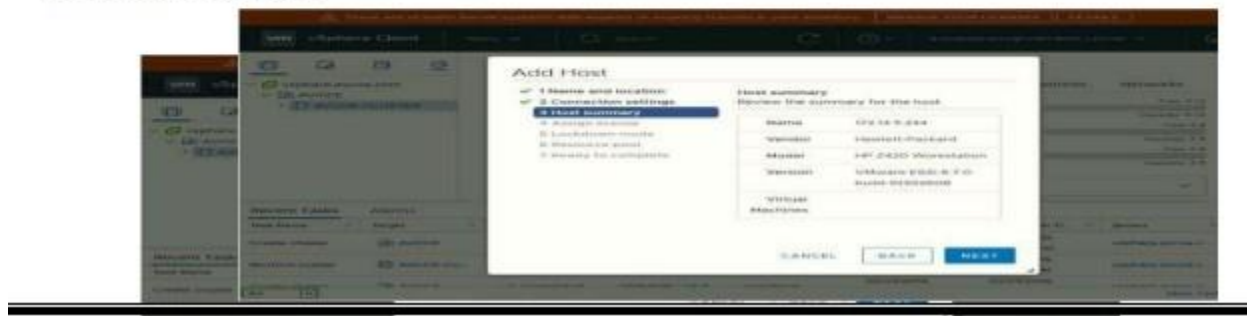
More T...

Configure...

Add host IP :



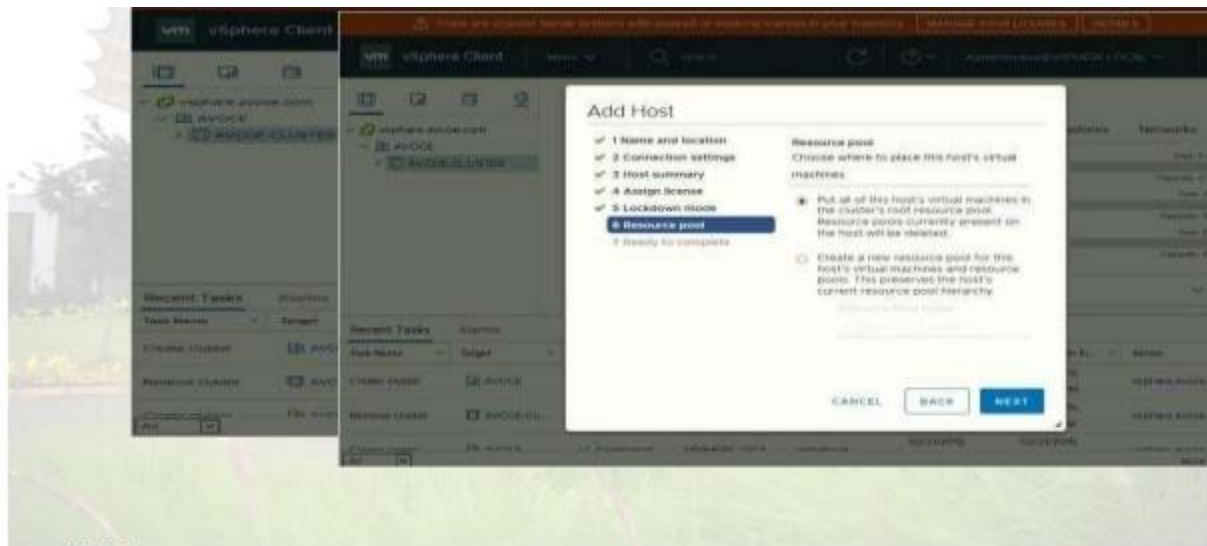
Enter host credential :



Host summary :

Lock Down mode:

Add Host In Pool:



Finish:



Host View and View Config:

Cluster View and Configuration:

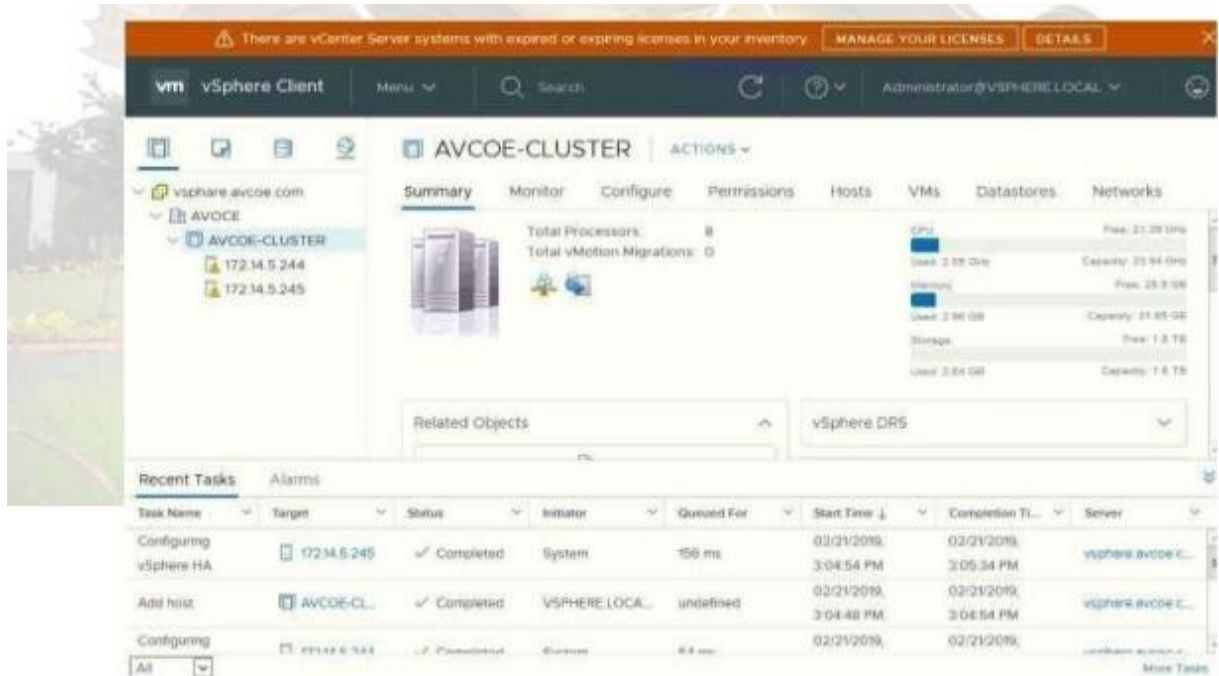
The image displays two screenshots of the vSphere Client interface, illustrating the Host View and Cluster View configurations.

Top Screenshot: Host View (172.14.5.245)

The interface shows the Host View for the host 172.14.5.245. The left sidebar lists the hierarchy: vSphere Client > vsphere.avcoe.com > AVCOE > AVCOE-CLUSTER > 172.14.5.244 > 172.14.5.245. The main pane displays the Summary tab for the host, showing details such as Hypervisor (VMware ESX), Model (MT Z420 Workstation), Processor Type (Intel(R) Xeon(R) CPU E5-1607 v2 @ 3.00GHz), Logical Processors (4), NICs (1), Virtual Machines (0), State (Connected), and Uptime (0 seconds). The right pane shows resource usage for CPU, Memory, and Storage.

Bottom Screenshot: Cluster View (AVCOE-CLUSTER)

The interface shows the Cluster View for the AVCOE-CLUSTER. The left sidebar lists the hierarchy: vSphere Client > vsphere.avcoe.com > AVCOE > AVCOE-CLUSTER > 172.14.5.244 > 172.14.5.245. The main pane displays the Summary tab for the cluster, showing details such as Total Processors (8) and Total vMotion Migrations (0). The right pane shows resource usage for CPU, Memory, and Storage. The bottom pane shows Related Objects, including vSphere DRS.



Conclusion: Like this we have configure VSphere Private Cloud